

Lecture 06

Matrix Operation II: Block Partition and Partitioned Product

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Announcements

Announcement in BB:

****3-Time Absence Rule:****

You are allowed to miss up to 3 classes during the semester without penalty.

****Grading Scheme for Attendance: ****

If you attend X, X-1, X-2, or X-3 classes (where X is the total number of classes requiring sign-in from 3rd week to the last one), you will receive the full 10% for attendance; If you attend X-4 classes, your attendance score will be calculated as $10\% \times (X-4)/(X-3)$.

Assignment 1 Available on BB

Today's Lecture: Outline

Today ... Continue Matrix Multiplication

Outline:

1. Block partition
2. Partitioned matrix multiplication

After this lecture, you should be able to

1. Conduct **multiplication of partitioned matrices**
2. **Especially**: Write the expressions of valid matrix-matrix multiplication

Recall: Matrix Times Matrix

Definition 5.1 (Matrix multiplication)

Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times k}$. The product $C = AB$ is an $m \times k$ matrix

where each entry is given by:
$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

Rows * columns

where a_{ik} is the (i, k) -th entry of A and b_{kj} is the (k, j) -th entry of B .

Remark: AB exists iff # of columns of A equal to # of rows of B .

The dimensions
have to match!

Property

Let $A \in \mathbb{R}^{m \times n}$, $B = [\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_r] \in \mathbb{R}^{n \times r}$. The product AB can be computed by:

$$AB = [A\mathbf{b}_1, A\mathbf{b}_2, \dots, A\mathbf{b}_r].$$

Valid Vector-vector products (**different size**)

Want to multiply $u \in \mathbb{R}^{m \times 1}$ and $v \in \mathbb{R}^{n \times 1}$, **two column vectors**.

You are allowed to use transpose.

Which products are valid?

In-class exercise: 1-2 min.

$$uv$$

Invalid

$$uv^T$$

Valid

$$u^T v$$

Invalid

$$vu$$

Invalid

$$v^T u$$

Invalid

$$vu^T$$

Valid

Part 1 Block Partition and Matrix Multiplication

25 mins

Matrix Partition: Existing Examples

Sometimes, it will often be convenient to think about matrices defined in terms of other matrices.

For example, we already saw augmented matrices, defined in terms of a coefficient matrix and a vector of righthand sides.

$$[A \mid \mathbf{b}].$$

Another example: express a matrix by vectors

$$\begin{bmatrix} \mathbf{a}^{(1)} \\ \mathbf{a}^{(2)} \\ \vdots \\ \mathbf{a}^{(m)} \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \dots & \mathbf{a}_n \end{bmatrix}$$

Example 1

Let's see a different example, with similar flavor of “partition”.

$$P = \begin{bmatrix} -1 & -1 & -1 \\ 2 & 2 & 2 \\ -1 & -1 & -1 \\ 2 & 2 & 2 \end{bmatrix}$$

What do you observe?

Can you simplify the expression?

Example 1

Let's see a different example, with similar flavor of “partition”.

$$P = \begin{bmatrix} -1 & -1 & -1 \\ 2 & 2 & 2 \\ -1 & -1 & -1 \\ 2 & 2 & 2 \end{bmatrix}$$

What do you observe?

Can you simplify the expression?

$$\text{Let } \mathbf{a} = [-1, -1, -1], \quad \mathbf{b} = [2, 2, 2] \quad B = \begin{bmatrix} \mathbf{a} \\ \mathbf{b} \end{bmatrix}$$

$$\text{Then } P = \begin{bmatrix} B \\ B \end{bmatrix}$$

Example 2

$$A = \left[\begin{array}{cc|cc|cc} 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ \hline 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \end{array} \right]$$

Can you simplify the expression?

4 by 6 matrix

2 by 2 blocks give

2 by 3 block matrix

Example 2

$$A = \left[\begin{array}{cc|cc|cc} 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ \hline 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \end{array} \right]$$

Can you simplify the expression?

4 by 6 matrix
2 by 2 blocks give
2 by 3 block matrix

$$\text{Let } I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \text{ then } A = \begin{bmatrix} I_2 & I_2 & I_2 \\ I_2 & I_2 & I_2 \end{bmatrix}$$

$$\text{Let } J = \begin{bmatrix} I_2 \\ I_2 \end{bmatrix}, \text{ then } A = \begin{bmatrix} J & J & J \end{bmatrix}$$

Benefits of Using “Block”

Benefits:

- 1) Save space.
- 2) Clear structure.
 - help reduce mistake.

Useful in **parallel computing**, **numerical methods**, and **structured matrices**.

E.g.

$$A = \left[\begin{array}{cc|cc|cc} 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ \hline 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \end{array} \right] \quad \hat{A} = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}$$

Are the two matrices the same?

Definition

The matrix

$$A = \begin{bmatrix} A_{11} & \cdots & A_{1t} \\ \vdots & \ddots & \vdots \\ A_{s1} & \cdots & A_{st} \end{bmatrix}$$

is a partition of matrix with $s \times t$ blocks if the matrices A_{ij} satisfies

- (1) For each fixed i , the number of rows of all A_{ij} are equal.
- (2) For each fixed j , the number of columns of all A_{ij} are equal.

The matrix A_{ij} is called the (i,j) -block of A .

We say A has s block-rows and t block-columns.

Make sure the blocks are the right sizes:

- (i) blocks in the same (block) row have same # of rows,
- (ii) blocks in the same (block) column need to have same # of columns.

Example

If

$$A_{11} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad A_{12} = \begin{bmatrix} 5 & -1 & 3 \\ -2 & 1 & 0 \end{bmatrix}, \quad A_{13} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$$

$$A_{21} = [1 \quad 5], \quad A_{22} = [7 \quad -2 \quad 3], \quad A_{23} = [2]$$

then

$$A = \left[\begin{array}{cc|ccc|c} 1 & 2 & 5 & -1 & 3 & 4 \\ 3 & 4 & -2 & 1 & 0 & 6 \\ \hline 1 & 5 & 7 & -2 & 3 & 2 \end{array} \right]$$

has the (1,2)-block A_{12} and (2,3)-block A_{23} . Moreover, the number of rows of all A_{1j} is 2, and the number of columns of all A_{i3} is 1.

The number of block-rows is 2

The number of block-columns is 3.

Non-examples

Non-example:

$$\left[- \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 7 & -2 & 3 \end{bmatrix} \right]$$

Non-example:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = \begin{bmatrix} A_1 & A_2 & A_3 \end{bmatrix}$$

where

$$A_1 = \begin{bmatrix} 1 & 2 \end{bmatrix}$$
$$A_2 = \begin{bmatrix} 4 & 5 \end{bmatrix}$$
$$A_3 = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$

Multiplication of Partitioned Matrices

Partitioned matrices multiplication:

Rule 1 (block structure match)

The number of block-columns in A
must equal the number of block-rows in B.

Then can treat each block as an “entry”,
for standard matrix multiplication. Level-1 multiplication

Rule 2 (block size match)

When multiplying two blocks, Level-2 multiplication
the dimensions of the blocks must match.

Multiplication of Partitioned Matrices

Example

Given partitioned matrices:

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \quad B = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$$

Step 1 (check Rule 1)

A has 2 block columns, matches 2 block rows of B.

Step 2 Compute candidate result (not sure it's valid yet).

$$AB \stackrel{\text{candidate}}{=} \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{bmatrix} \quad \text{Level-1 multiplication}$$

Step 3 (check Rule 2): There are 8 products $A_{ij}B_{jk}$. Level-2 multiplication

For them to be valid, need # of columns of A_{ij} = # of rows of B_{jk} , for any i, j, k .

[see next page for details]

Rule 2: Checking What?

$$\begin{array}{cc} p_1 & p_2 \\ r_1 & \begin{bmatrix} A_{11} & A_{12} \end{bmatrix} \\ r_2 & \begin{bmatrix} A_{21} & A_{22} \end{bmatrix} \end{array} \qquad \begin{array}{cc} c_1 & c_2 \\ q_1 & \begin{bmatrix} B_{11} & B_{12} \end{bmatrix} \\ q_2 & \begin{bmatrix} B_{21} & B_{22} \end{bmatrix} \end{array}$$

In-class Exercise: (2-3 min)

Write the dim-checking condition for each product to be valid.

$$AB = \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{bmatrix}$$

$r_1 \times p_1$ $q_1 \times c_1$
 $r_1 \times p_1$ $q_1 \times c_1$
 For $A_{11} B_{11}$ to be valid,
 what's needed?
 Restate: $A_{11} B_{11}$, for $A_{11} B_{11}$ to be valid.
 $r_1 \times p_1$ $q_1 \times c_1$
 require? Answer: $p_1 = q_1$

How Many Product to Be Checked?

Claim: If A_{11} 's # of columns = B_{11} 's # of rows,
 A_{12} 's # of columns = B_{21} 's # of rows,
Then Rule 2 is satisfied.

$$\begin{array}{cc} p_1 & p_2 \\ r_1 & \begin{bmatrix} A_{11} & A_{12} \end{bmatrix} \\ r_2 & \begin{bmatrix} A_{21} & A_{22} \end{bmatrix} \end{array} \quad \begin{array}{cc} c_1 & c_2 \\ q_1 & \begin{bmatrix} B_{11} & B_{12} \end{bmatrix} \\ q_2 & \begin{bmatrix} B_{21} & B_{22} \end{bmatrix} \end{array}$$

$$AB = \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{bmatrix}$$

If $p_1 = q_1$ $p_2 = q_2$ then all products are valid.

Part 2 Matrix-matrix products

Strang's book Sec 2.4

30-35 mins

Remark of Notation: Rows With & Without Transpose

Consider $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$

Notation 1:

Here

$$\mathbf{A} = \begin{bmatrix} a^{(1)} \\ a^{(2)} \end{bmatrix} \quad \begin{array}{l} a^{(1)} = [1, 2, 3] \\ a^{(2)} = [4, 5, 6] \end{array}$$

Notation 2:

Here

$$\mathbf{A} = \begin{bmatrix} a_{(1)}^\top \\ a_{(2)}^\top \end{bmatrix} \quad \begin{array}{l} a_{(1)}^\top = [1, 2, 3] \\ a_{(2)}^\top = [4, 5, 6] \end{array} \quad \begin{array}{l} \mathbf{a}_{(1)} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \\ \mathbf{a}_{(2)} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} \end{array}$$

From now on, we will mostly use “column vectors”, for convenience.
Will explain the benefits later.

Benefits of Using Column Vectors

Two forms: $\mathbf{A} = \begin{bmatrix} a^{(1)} \\ a^{(2)} \end{bmatrix}$ $\mathbf{A} = \begin{bmatrix} a_{(1)}^\top \\ a_{(2)}^\top \end{bmatrix}$

Recommend the 2nd form. Using “column vectors” from now on.

Benefit: It is easy to memorize what products are valid.

Consider column vector u (size n), v (size n), and matrix A (size $n \times n$).

For each of the following: valid or invalid?

Memorize the results!

Vector times vector: uv ,
✗

uv^T ,
✓

$v^T u$,
✓

Matrix times vector: Av ,
 $n \times n$ $n \times 1$
✓

Av^T ,
 $n \times n$ $1 \times n$
✗

vA ,
 $n \times 1$ $n \times n$
✗

$v^T A$,
 $1 \times n$ $n \times n$
✓

Exercise: What are Valid?

Exercise during Class Break

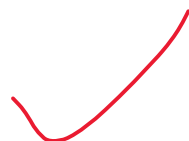
Consider column vector u (size n), v (size n),
and matrix A, B, C (size $n \times n$).

Beginner: check dim.
Expert: check in 10 sec.

$$uAv$$

uA not valid



$$u^T Av$$


$$uAvuBCv$$

not valid not valid



$$u^T Avu^T BCv$$


Multiplication of Partitioned Matrices

What if we multiply A and B?

Q1: What pair of partitions lead to valid multiplication?

Q2: What results do we get?

A: whole matrix, row-partition, column-partition

B: whole matrix, row-partition, column-partition

9 combinations.

Q1: Among them, which are valid?

For simplicity, consider

$$\mathbf{A} = [a_1, a_2, a_3] = \begin{bmatrix} a_{(1)}^\top \\ a_{(2)}^\top \end{bmatrix}, \quad \mathbf{A} \in \mathbb{R}^{2 \times 3}$$

$$\mathbf{B} = [b_1, b_2] = \begin{bmatrix} b_{(1)}^\top \\ b_{(2)}^\top \\ b_{(3)}^\top \end{bmatrix}, \quad \mathbf{B} \in \mathbb{R}^{3 \times 2}$$

Table of Multiplications: First Group

		1×1	1×2
		B	$\begin{bmatrix} b_{(1)}^\top \\ b_{(2)}^\top \\ b_{(3)}^\top \end{bmatrix}$ $[b_1, b_2]$
1×1	A	Yes	?
2×1	$\begin{bmatrix} a_{(1)}^\top \\ a_{(2)}^\top \end{bmatrix}$	?	
	$[a_1, a_2, a_3]$		

$$\mathbf{A} = [a_1, a_2, a_3] = \begin{bmatrix} a_{(1)}^\top \\ a_{(2)}^\top \end{bmatrix}, \quad \mathbf{A} \in \mathbb{R}^{2 \times 3}$$

$$\mathbf{B} = [b_1, b_2] = \begin{bmatrix} b_{(1)}^\top \\ b_{(2)}^\top \\ b_{(3)}^\top \end{bmatrix}, \quad \mathbf{B} \in \mathbb{R}^{3 \times 2}$$

Matrix * Column-form

Is $A \cdot [b_1, b_2]$ valid? $\mathbf{A} = [a_1, a_2, a_3] = \begin{bmatrix} a_{(1)}^\top \\ a_{(2)}^\top \end{bmatrix}, \quad \mathbf{A} \in \mathbb{R}^{2 \times 3} \quad \mathbf{B} = [b_1, b_2] = \begin{bmatrix} b_{(1)}^\top \\ b_{(2)}^\top \\ b_{(3)}^\top \end{bmatrix}, \quad \mathbf{B} \in \mathbb{R}^{3 \times 2}$

Step 1: Check Block-Structure Dimensions (Rule 1)

- A has 1 block-column, and $[b_1, b_2]$ has 1 block-row.
- Block structure matches.

Step 2: Compute Product Level-1 multiplication

$$A \cdot [b_1, b_2] = [Ab_1, Ab_2].$$

Step 3: Check Component Dimensions (Rule 2)

- Ensure Ab_1 and Ab_2 are valid. Level-2 multiplication
- A is 2×3 , b_1, b_2 are 3×1 , so multiplication is valid. ✓

Row-form * Matrix

Is $\begin{bmatrix} a_{(1)}^\top \\ a_{(2)}^\top \end{bmatrix} \cdot B$ valid?

$$\mathbf{A} = [a_1, a_2, a_3] = \begin{bmatrix} a_{(1)}^\top \\ a_{(2)}^\top \end{bmatrix}, \quad \mathbf{A} \in \mathbb{R}^{2 \times 3}$$

$$\mathbf{B} = [b_1, b_2] = \begin{bmatrix} b_{(1)}^\top \\ b_{(2)}^\top \\ b_{(3)}^\top \end{bmatrix}, \quad \mathbf{B} \in \mathbb{R}^{3 \times 2}$$

Step 1: Check block structure.

$\begin{bmatrix} a_{(1)}^\top \\ a_{(2)}^\top \end{bmatrix}$ has 1 block-column, B has 1 block-row. Match; continue.

Step 2: Compute products

Level-1 multiplication

$$\begin{bmatrix} a_{(1)}^\top \\ a_{(2)}^\top \end{bmatrix} \cdot B = \begin{bmatrix} a_{(1)}^\top B \\ a_{(2)}^\top B \end{bmatrix}$$

Each row $a_{(i)}^\top$ is multiplied with B separately.

Step 3: Check component dimensions (Rule 2)

- Each $a_{(i)}^\top$ is 1 x 3, B is 3 x 2
- Each product $a_{(i)}^\top B$ results in 1 x 2 row

Level-2 multiplication

Table of Multiplications: : Second Group

		3×1	1×2
	B	$\begin{bmatrix} b_{(1)}^\top \\ b_{(2)}^\top \\ b_{(3)}^\top \end{bmatrix}$	$[b_1, b_2]$
A	Yes		Yes
2×1 $\begin{bmatrix} a_{(1)}^\top \\ a_{(2)}^\top \end{bmatrix}$	Yes	?	
1×3 $[a_1, a_2, a_3]$			?

Table of Multiplications: : Second Group

		3×1	1×2
	B	$\begin{bmatrix} b_{(1)}^\top \\ b_{(2)}^\top \\ b_{(3)}^\top \end{bmatrix}$	$[b_1, b_2]$
A	Yes		Yes
2×1	$\begin{bmatrix} a_{(1)}^\top \\ a_{(2)}^\top \end{bmatrix}$	Yes	No
1×3	$[a_1, a_2, a_3]$		No

Diagram illustrating the multiplication table for the Second Group. The table is a 4x4 grid. The first row and column are headers for dimensions. The cells contain the result of the multiplication (Yes/No) and the resulting vector/matrix. Red arrows labeled "not match" point from the (A, B) cell to the (2x1, 3x1) cell and from the (A, B) cell to the (1x3, 1x2) cell. Blue arrows point from the (A, B) cell to the (2x1, 2x1) cell and from the (A, B) cell to the (1x3, 1x2) cell.

Table of Multiplications: : Third Group

	1×1	3×1	
	B	$\begin{bmatrix} b_{(1)}^\top \\ b_{(2)}^\top \\ b_{(3)}^\top \end{bmatrix}$	$[b_1, b_2]$
1×1 A	Yes	?	Yes
2×1 $\begin{bmatrix} a_{(1)}^\top \\ a_{(2)}^\top \end{bmatrix}$	Yes	No	
1×3 $[a_1, a_2, a_3]$	?		No

Column-form Times Matrix

Is $[a_1, a_2, a_3] B$ valid?

$$A = [a_1, a_2, a_3] \in \mathbb{R}^{2 \times 3}, \quad B \in \mathbb{R}^{3 \times 2}.$$

$$a_i \in \mathbb{R}^{2 \times 1} \quad (i = 1, 2, 3).$$

Step 1: Check block structure.

$[a_1, a_2, a_3]$ has 3 block-columns, but B has 1 block-row. Mismatch.

So the product is invalid. Level-1 multiplication

What if we try to expand the product anyways?

Column-form Times Matrix

Is $[a_1, a_2, a_3] B$ valid?

$$A = [a_1, a_2, a_3] \in \mathbb{R}^{2 \times 3}, \quad B \in \mathbb{R}^{3 \times 2}.$$

$$a_i \in \mathbb{R}^{2 \times 1} \quad (i = 1, 2, 3).$$

Step 1: Check block structure.

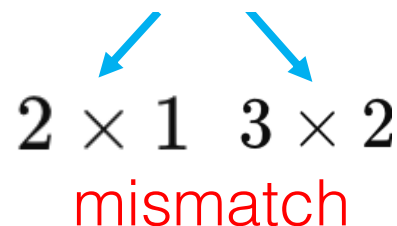
$[a_1, a_2, a_3]$ has 3 block-columns, but B has 1 block-row. Mismatch.

So the product is invalid.

What if we try to expand it anyway?

- Attempting $[a_1, a_2, a_3] \cdot b = [a_1 b, a_2 b, a_3 b]$ treats it as scalar-vector multiplication.

$$AB = [a_1 B, \quad a_2 B, \quad a_3 B]$$


$$\begin{array}{cc} 2 \times 1 & 3 \times 2 \\ \text{mismatch} \end{array}$$

Level-2 multiplication

Still an invalid expression.

Table of Multiplications: : Third Group

	1×1	3×1	
	B	$\begin{bmatrix} b_{(1)}^\top \\ b_{(2)}^\top \\ b_{(3)}^\top \end{bmatrix}$	$[b_1, b_2]$
1×1 A	Yes	No	Yes
$\begin{bmatrix} a_{(1)}^\top \\ a_{(2)}^\top \end{bmatrix}$	Yes	No	
1×3 $[a_1, a_2, a_3]$	No		No

Table of Multiplications: Fourth Group

	B	3×1 $\begin{bmatrix} b_{(1)}^\top \\ b_{(2)}^\top \\ b_{(3)}^\top \end{bmatrix}$	1×2 $[b_1, b_2]$
A	Yes	No	Yes
2×1 $\begin{bmatrix} a_{(1)}^\top \\ a_{(2)}^\top \end{bmatrix}$	Yes	No	?
1×3 $[a_1, a_2, a_3]$	No	?	No

Red lines and handwritten notes indicating matches:

- A red line connects the 2×1 dimension of A to the 3×1 dimension of B, labeled "match".
- A red line connects the 1×3 dimension of A to the 1×2 dimension of B, labeled "match".

Column-form * Row-form

Is $[a_1, a_2, a_3] \begin{bmatrix} b_{(1)}^\top \\ b_{(2)}^\top \\ b_{(3)}^\top \end{bmatrix}$ valid? $A = [a_1, a_2, a_3] \in \mathbb{R}^{2 \times 3}$, $B = \begin{bmatrix} b_{(1)}^\top \\ b_{(2)}^\top \\ b_{(3)}^\top \end{bmatrix} \in \mathbb{R}^{3 \times 2}$.

Step 1: Check Block-Structure Dimensions (Rule 1)

- $[a_1, a_2, a_3]$ has 3 block-columns.
- $\begin{bmatrix} b_{(1)}^\top \\ b_{(2)}^\top \\ b_{(3)}^\top \end{bmatrix}$ has 3 block-rows.

Step 2: Compute Product Level-1 multiplication

$$[a_1, a_2, a_3] \begin{bmatrix} b_{(1)}^\top \\ b_{(2)}^\top \\ b_{(3)}^\top \end{bmatrix} = a_1 b_{(1)}^\top + a_2 b_{(2)}^\top + a_3 b_{(3)}^\top$$

Step 3: Check component dimensions (Rule 2)

- Each a_i is 2×1 , each $b_{(j)}^\top$ is 1×2
- Each product results in 2×2 matrix. The final result is a 2×2 matrix.

Level-2 multiplication

Rows * Columns & Columns * Rows

Form 1	$\begin{bmatrix} a_{(1)}^\top \\ \vdots \\ a_{(m)}^\top \end{bmatrix}$	$[b_1 \quad b_2 \quad \cdots \quad b_k]$	$(\underbrace{a_{(i)}^\top}_{\text{row}} \underbrace{b_j}_{\text{col}})_{m \times k}$	Definition
Form 2	$[a_1 \quad \cdots \quad a_n]$	$\begin{bmatrix} b_{(1)}^\top \\ \vdots \\ b_{(n)}^\top \end{bmatrix}$	$\sum_{i=1}^n \underbrace{a_i}_{\text{col}} \underbrace{b_{(i)}^\top}_{\text{row}}$	sum of outer pdt

Rows times columns:

$$\begin{bmatrix} a_{(1)}^\top \\ \vdots \\ a_{(m)}^\top \end{bmatrix} [b_1, \dots, b_k] = \begin{bmatrix} a_{(1)}^\top b_1 & \cdots & a_{(1)}^\top b_k \\ \vdots & \ddots & \vdots \\ a_{(m)}^\top b_1 & \cdots & a_{(m)}^\top b_k \end{bmatrix}$$

Columns times rows:

$$[a_1, \dots, a_n] \begin{bmatrix} b_{(1)}^\top \\ \vdots \\ b_{(n)}^\top \end{bmatrix} = [a_1 b_{(1)}^\top + \cdots + a_n b_{(n)}^\top]$$

Observation:

Form 1:

Level-1: outer product,

Level-2: consists of inner product

Form 2:

level-1: inner product,

Level 2: consists of outer product

Rows * Columns & Columns * Rows: Example

Rows times columns:

$$\begin{bmatrix} a_{(1)}^\top \\ \vdots \\ a_{(m)}^\top \end{bmatrix} [b_1, \dots, b_k] = \begin{bmatrix} a_{(1)}^\top b_1 & \cdots & a_{(1)}^\top b_k \\ \vdots & \ddots & \vdots \\ a_{(m)}^\top b_1 & \cdots & a_{(m)}^\top b_k \end{bmatrix}$$

Columns times rows:

$$[a_1, \dots, a_n] \begin{bmatrix} b_{(1)}^\top \\ \vdots \\ b_{(n)}^\top \end{bmatrix} = [a_1 b_{(1)}^\top + \cdots + a_n b_{(n)}^\top]$$

Example of columns times rows:

$$\begin{bmatrix} 1 & 4 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} [3, 2] + \begin{bmatrix} 4 \\ 5 \end{bmatrix} [1, 0] = \begin{bmatrix} 1 \times 3 & 1 \times 2 \\ 1 \times 3 & 1 \times 2 \end{bmatrix} + \begin{bmatrix} 4 \times 1 & 4 \times 0 \\ 5 \times 1 & 5 \times 0 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 3 & 2 \end{bmatrix} + \begin{bmatrix} 4 & 0 \\ 5 & 0 \end{bmatrix} = \begin{bmatrix} 7 & 2 \\ 8 & 2 \end{bmatrix}$$

Example of rows times column (definition):

$$\begin{bmatrix} 1 & 4 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 \times 3 + 4 \times 1 & 1 \times 2 + 4 \times 0 \\ 1 \times 3 + 5 \times 1 & 1 \times 2 + 5 \times 0 \end{bmatrix} = \begin{bmatrix} 7 & 2 \\ 8 & 2 \end{bmatrix}$$

Rules	Computation in Example	Interpretation of product
rows * columns	4 dot products	Collection of inner products
columns*rows	2 outer products	Sum of outer products

Example 2: Sum of Outer Products

Example

Given

$$X = \begin{bmatrix} 3 & 1 \\ 2 & 4 \\ 1 & 2 \end{bmatrix}, \quad Y = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 1 \end{bmatrix}$$

Compute XY

$$\begin{aligned} XY &= \left[\begin{array}{c|c} 3 & 1 \\ 2 & 4 \\ 1 & 2 \end{array} \right] \left[\begin{array}{ccc} 1 & 2 & 3 \\ 2 & 4 & 1 \end{array} \right] = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} + \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix} \begin{bmatrix} 2 & 4 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 3 & 6 & 9 \\ 2 & 4 & 6 \\ 1 & 2 & 3 \end{bmatrix} + \begin{bmatrix} 2 & 4 & 1 \\ 8 & 16 & 4 \\ 4 & 8 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 5 & 10 & 10 \\ 10 & 20 & 10 \\ 5 & 10 & 5 \end{bmatrix} \end{aligned}$$

Table of Multiplications: Fourth Group

		3×1 $\begin{bmatrix} b_{(1)}^\top \\ b_{(2)}^\top \\ b_{(3)}^\top \end{bmatrix}$	1×2 $[b_1, b_2]$
	B		
A	Yes	No	Yes
2×1 $\begin{bmatrix} a_{(1)}^\top \\ a_{(2)}^\top \end{bmatrix}$	Yes	No	Yes
1×3 $[a_1, a_2, a_3]$	No	Yes	No

Multiplication of Partitioned Matrices

What if we multiply A and B?

Q1: What partitions are valid? Q2: What results do we get?

A: whole-matrix, row-partition, column-partition
B: whole-matrix, row-partition, column-partition

$$A: m \times n; \quad B: n \times k.$$

Level-1 multiplication	Form of A	Form of B	Product	Remark
scalar * scalar	A	B	AB	
scalar * vector	A	$[\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_k]$	$[A\mathbf{b}_1, A\mathbf{b}_2, \dots, A\mathbf{b}_k]$	
vector * scalar	$\begin{bmatrix} a_{(1)}^\top \\ \vdots \\ a_{(m)}^\top \end{bmatrix}$	B	$\begin{bmatrix} a_{(1)}^\top B \\ \vdots \\ a_{(m)}^\top B \end{bmatrix}$	
Vector outer product	$\begin{bmatrix} a_{(1)}^\top \\ \vdots \\ a_{(m)}^\top \end{bmatrix}$	$[\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_k]$	$(\mathbf{a}_{(i)}^\top \mathbf{b}_j)_{m \times k}$	Definition
Vector inner product	$[\mathbf{a}_1, \dots, \mathbf{a}_n]$	$\begin{bmatrix} b_{(1)}^\top \\ \vdots \\ b_{(n)}^\top \end{bmatrix}$	$\sum_{i=1}^n \mathbf{a}_i \mathbf{b}_{(i)}^\top$	Sum of outer-products

Practice Question: When is each form valid?

Where does “**dim-checking**” play a role?

Part 3 # of Operations in Matrix Multiplication

10 mins

How Many Operations: Vector-vector products

Suppose u, v are two vectors.

To compute $u^T v$, how many operations are needed?

$$u = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}, \quad v = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \quad u^T v = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$$

n mults + $(n-1)$ adds $\approx 2n$

Example

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}^T \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = 1 \cdot 4 + 2 \cdot 5 + 3 \cdot 6$$

3 multiplications
2 additions
5 total operations

How Many Operations: Matrix-vector products

To compute matrix times vector: how many operations?

$$A \in \mathbb{R}^{m \times n}, \quad x \in \mathbb{R}^{n \times 1} \quad y = Ax \in \mathbb{R}^{m \times 1}$$

Use row-form:

$$A = \begin{bmatrix} a_1^\top \\ a_2^\top \\ \vdots \\ a_m^\top \end{bmatrix}, \quad a_i^\top \in \mathbb{R}^{1 \times n}. \quad y = Ax = \begin{bmatrix} a_1^\top x \\ a_2^\top x \\ \vdots \\ a_m^\top x \end{bmatrix}$$

We compute m vector-vector (size n) products.

Each product requires $\sim 2n$ operations.

In total, $\sim 2mn$ operations.

How Many Operations: Matrix-matrix products

To compute matrix times matrix: how many operations?

$$A \in \mathbb{R}^{m \times n}, \quad B \in \mathbb{R}^{n \times p} \quad C = AB \in \mathbb{R}^{m \times p}$$

Use definition:

$$c_{ij} = (\text{row } i \text{ of } A) \cdot (\text{column } j \text{ of } B)$$

We compute mp vector-vector (size n) products.

Each product requires $\sim 2n$ operations.

In total, $\sim 2mnp$ operations.

How Many Operations

Summary of Operations: General matrices

Operation	Input sizes	Output size	Approx. # of operations (mults + adds) 
Vector–vector $u^\top v$	$u, v \in \mathbb{R}^{n \times 1}$	scalar $\in \mathbb{R}$	$\approx 2n$
Matrix–vector Ax	$A \in \mathbb{R}^{m \times n}, x \in \mathbb{R}^{n \times 1}$	$y \in \mathbb{R}^{m \times 1}$	$\approx 2mn$
Matrix–matrix AB	$A \in \mathbb{R}^{m \times n}, B \in \mathbb{R}^{n \times p}$	$C \in \mathbb{R}^{m \times p}$	$\approx 2mnp$

Simple case: square matrices

Operation	Sizes (square case)	Output size	Approx. # of operations (mults + adds)
Vector–vector $u^\top v$	$u, v \in \mathbb{R}^{n \times 1}$	scalar $\in \mathbb{R}$	$\approx 2n$
Matrix–vector Ax	$A \in \mathbb{R}^{n \times n}, x \in \mathbb{R}^{n \times 1}$	$y \in \mathbb{R}^{n \times 1}$	$\approx 2n^2$
Matrix–matrix AB	$A, B \in \mathbb{R}^{n \times n}$	$C \in \mathbb{R}^{n \times n}$	$\approx 2n^3$

Memorize
these three
results!

Concluding Section

Short summary of the past few lectures

Lec 03: matrix-vector multiplication

Represent $Ax = b$

Lec 05: matrix-matrix multiplication
(+ transpose)

Will be useful later for
representing GE

Lec 06:

Row/column-partitioned matrix multiplication;
—matrix as sum of outer-products

Useful in SVD
(~ lec 23)



Summary Today

Write your summary below.

One sentence summary:

Detailed summary:

Summary Today

Instructor's summary

One sentence summary:

We learned (deeper) matrix multiplication, transpose and partition.

Detailed summary:

—Block partition:

Rules of partition and multiplication.

—Application: Valid partitioned matrix-matrix multiplication. (row-form and column-forms)

Valid forms:

F1: (rows) $\times$ (columns) Definition!

F2: (matrix) $\times$ (columns)

F3: (rows) $\times$ (matrix)

F4: (columns) $\times$ (rows) = sum of outer products